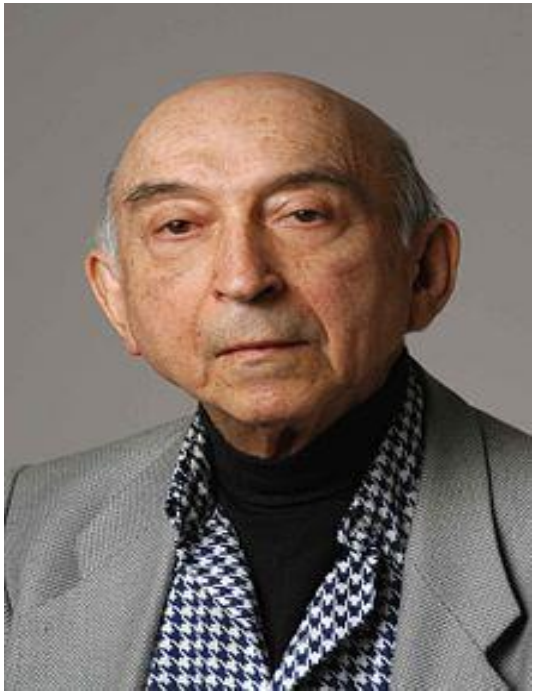
Fuzzy Logic In AI

Prof. Dr. A K M Akhtar Hossain
Dept. of CSE, RU

Fuzzy Logic History

- The term *fuzzy logic* was introduced by **Lotfi A. Zadeh** in the year **1965.**



Lotfi Aliasker Zadeh was born in 4 February 1921 in Baku, Azerbaijan, SSR.

What is Fuzzy Logic?

Fuzzy Logic is a type of logic where something can be partly true and partly false.

- **Fuzzy logic** is a form of **many-valued logic** in which the truth values of variables may be any real number between 0 and 1, considered to be "fuzzy".
- In Boolean logic, the truth values of variables may only be 0 or 1, often called "crisp" values.
- **Fuzzy logic has been employed to handle the concept of partial truth, where the truth value may range between completely true and completely false.**
- Furthermore, when linguistic variables are used, these degrees may be managed by specific (membership) functions.

Fuzzy logic

- Fuzzy logic is a branch of fuzzy set theory, which deals with the representation and inference from knowledge. Fuzzy logic, unlike other logical systems, deals with imprecise or uncertain knowledge.
- In this narrow, and perhaps correct sense, fuzzy logic is just one of the branches of fuzzy set theory.

Fuzzy Applications

- Pattern recognition and classification
- Fuzzy clustering
- Image and speech processing
- Fuzzy systems for prediction
- Fuzzy control
- Monitoring
- Diagnosis
- Optimization and decision making
- Group decision making
- **Washing machines**
- **Electric Fan**; etc.

Applications

- **Washing machines** now use fuzzy logic. They can sense how heavy a load is and determine the correct amount of water and detergent, speed of agitation, and length of the wash cycles.
- There is no single standard for the dirty laundry. Fuzzy logic enables the machine's computer to make “in between” decisions.

Example:

- **Words like young, tall, good, or high are fuzzy.**

❑ There is no single quantitative value which defines the term **young**.

❑ For some people, age **25 is young**, and for others, **age 35 is young.**

❑ The concept young has no clean boundary.

❑ Age 1 is definitely **child** and age 100 is definitely **not child**;

❑ Age 35 has some possibility of being young and usually depends.

- **On the context in which it is being considered.**

Examples: Fuzzy Linguistic variables

- Fuzzy sets and concepts are commonly used in natural language :-
- John is tall.
- Dan is smart.
- Alex is happy .
- Today is hot .
- Car runs faster.

Fuzzy set theory

- Fuzzy set theory is an extension of classical set theory where elements have varying degrees of membership. Membership function = converts an input into a value between 0 and 1
- A logic based on the two truth values, *True and False, is sometimes inadequate when* describing human reasoning.
- Fuzzy logic uses the whole interval between **0 (false) and 1 (true) to describe human reasoning.**

A **Fuzzy Set**

- A **Fuzzy Set is any set that allows its members to have different** degree of membership, called **membership function*, in the interval***

  [0 , 1].

- The **degree of membership or truth is not same as probability;**

❑ fuzzy truth is not likelihood of some event or condition.

❑ fuzzy truth represents membership in vaguely defined sets;

Fuzzy logic

- **Fuzzy logic is derived from fuzzy set theory dealing with reasoning** that is approximate rather than precisely deduced from classical predicate logic.
- Fuzzy logic is capable of handling inherently imprecise concepts.
- Fuzzy logic allows in linguistic form the set membership values to imprecise concepts like **"slightly", "quite" and "very".**
- Fuzzy set theory defines Fuzzy Operators on Fuzzy Sets.

Crisp and Non-Crisp Set

The notations used to express these mathematically are

$$A : X \rightarrow [0, 1]$$

$$A(x) = 1, \; x \text{ is a member of } A$$

$$A(x) = 0, \; x \text{ is not a member of } A$$

} Eq.(1)

A crisp set has a strict boundary.
An item is either:
0 Not a member
1 Member

A non-crisp (fuzzy) set does not have a strict boundary.
An item can belong to a set partially.
Membership can be any value between 0 and 1.

Crisp and Non-Crisp Set

Alternatively, the set **A** can be represented for all elements $x \in X$ by its characteristic function $\mu_A(x)$ defined as

$$\mu_A(x) = \begin{cases} 1 & \text{if } x \in X \\ 0 & \text{otherwise} \end{cases} \qquad \text{Eq.(2)}$$

Thus in classical set theory $\mu_A(x)$ has only the values **0** ('false') and **1** ('true''). Such sets are called **crisp sets.**

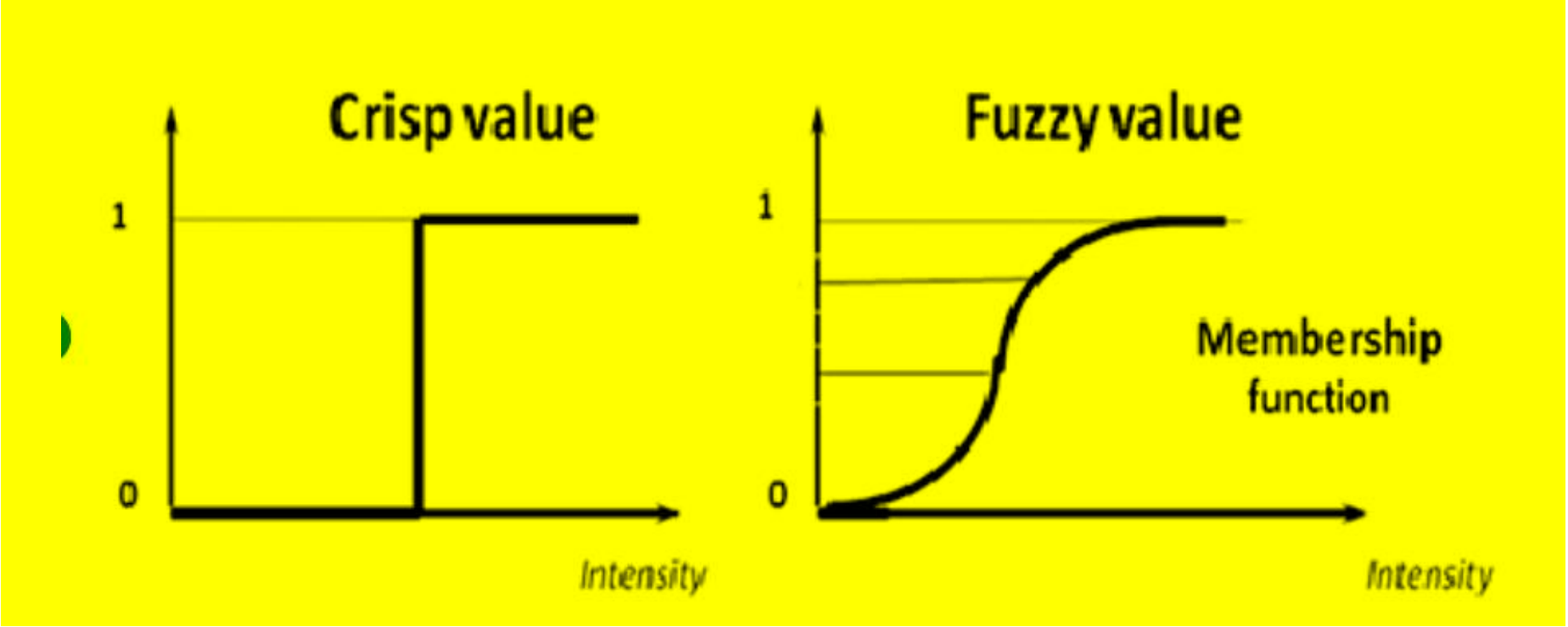
Non-Crisp Set

- For Non-crisp sets the characteristic function $\mu_A(x)$ can be defined.
  - The characteristic function $\mu_A(x)$ of Eq. (2) for the crisp set is generalized for the Non-crisp sets.
  - This generalized characteristic function $\mu_A(x)$ of Eq.(2) is called **membership function**.

  Such Non-crisp sets are called **Fuzzy Sets**.

- Crisp set theory is not capable of representing descriptions and classifications in many cases; In fact, Crisp set does not provide adequate representation for most cases.

Crisp and Non-Crisp Value



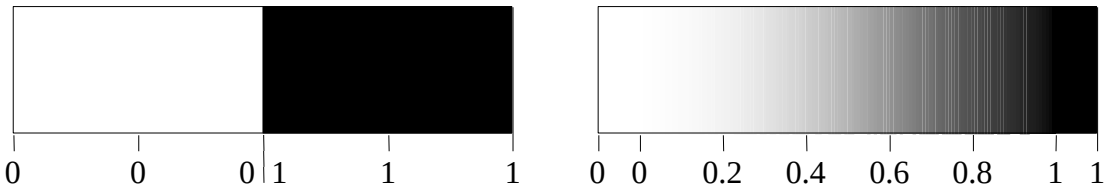
Difference between Crisp and Fuzzy Sets.

More Definitions (On Fuzzy Logic)

- Fuzzy logic is a set of mathematical principles for knowledge representation based on **degrees of membership**.

- Unlike two-valued Boolean logic, fuzzy logic is **multi-valued**. It deals with **degrees of membership** and **degrees of truth**.

- Fuzzy logic uses the continuum of logical values between 0 (completely false) and 1 (completely true). Instead of just black and white, it employs the spectrum of colours, accepting that things can be partly true and partly false at the same time.

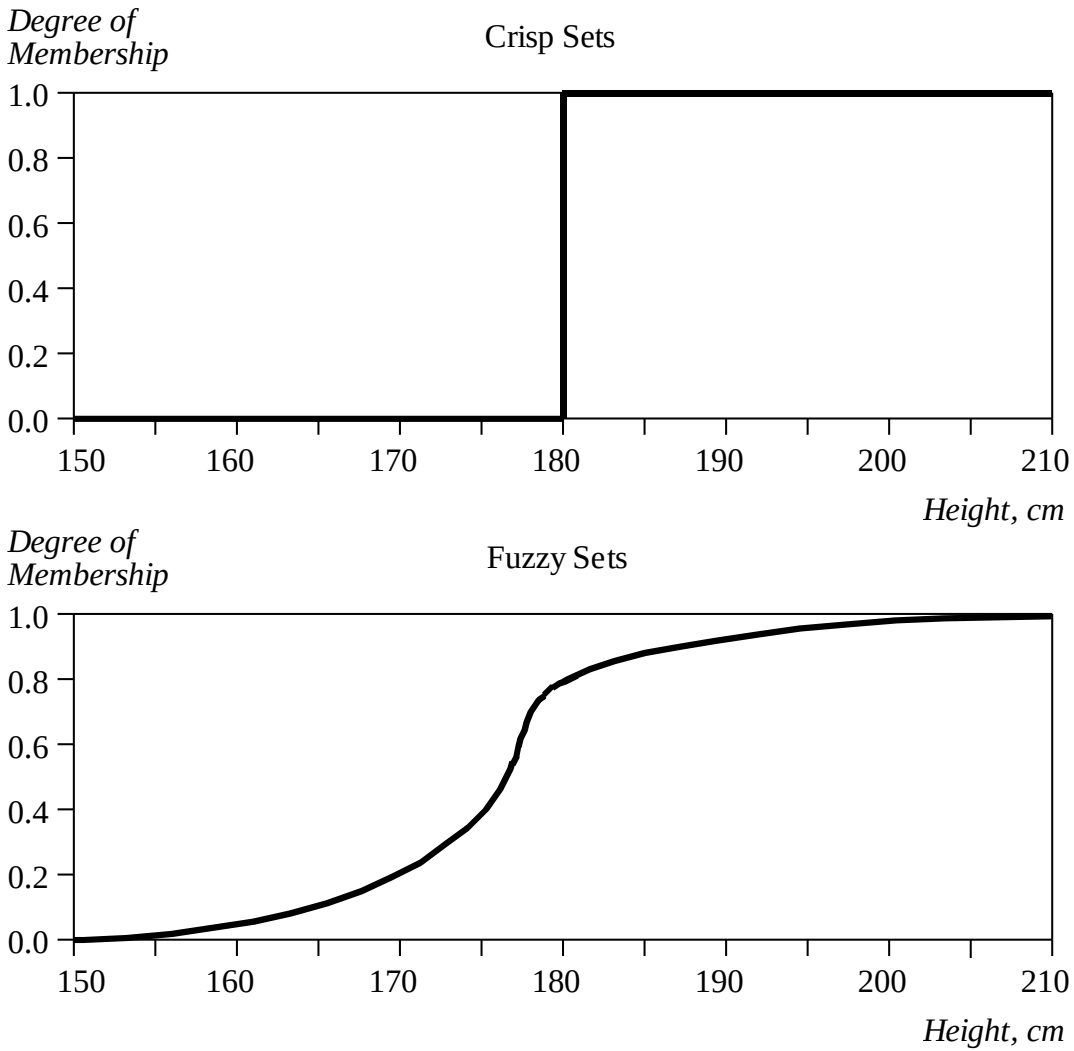


(*a*) Boolean Logic. (*b*) Multi-valued Logic.

Crisp Vs Fuzzy Sets

The x-axis represents the **universe of discourse** – the range of all possible values applicable to a chosen variable. In our case, the variable is the man height. According to this representation, the universe of men's heights consists of all tall men.

The y-axis represents the **membership value of the fuzzy set**. In our case, the fuzzy set of "tall men" maps height values into corresponding membership values.



Example: Fuzzy Sets

- **Hard science with IF-THEN rules:-**
- Example:- The logic for a simple temperature regulator that uses a fan might look like this:-

Example:

- IF temperature is **very cold** THEN **stop fan**
- IF temperature is **cold** THEN fan speed is **zero**
- IF temperature is **warm** THEN fan speed is **moderate**
- IF temperature is **hot** THEN fan speed is **high**

© Prof. Dr. A K M Akhtar Hossain
Dept. of CSE, RU

- **THE END**
- **Thanks**